

- Let  $\mathbb{Z}_n := \{0, 2, \dots, n-1\}$  and define

$$U_n := \left\{ \exp\left(\frac{2k\pi i}{n}\right) : k \in \mathbb{Z}_n \right\},$$

then  $(U_n, \cdot)$  is the  $n$ -th root of unity group.

- Cancellation laws:**  $\forall a, b, u, v \in G$ , if  $au = av$ , then  $u = v$ ; if  $ub = vb$ , then  $u = v$ .
- $\text{GL}_n(\mathbb{F}) := \{A \in \mathcal{M}_{n \times n} : \det(A) \neq 0\}$ .
- $\forall$  finite  $\Sigma$  with  $|\Sigma| = n$ , define  $S_{n, \Sigma}$  to be all bijections from  $\mathbb{Z}_n^+$  to  $\Sigma$ , then  $|S_{n, \Sigma}| = n!$ . (Induction with disjoint union on image of  $k+1$ .)  $\implies |S_n| = n!$ .
- $S_n$  is non-Abelian iff  $n \geq 3$  ( $\implies : \exists ab \neq ba$  but neither is identity;  $\Leftarrow : (12)$  and  $(13)$  don't commute)
- Dihedral:**  $r^n = s^2 = 1, rs = sr^{-1}$ .
- $S_n$  is generated by transpositions (induction, define  $\tau_\sigma$  to swap  $k+1$  and  $\sigma(k+1)$  if they are unequal and id otherwise.  $\tau_\sigma \circ \sigma \in S_k$  and  $\tau_\sigma = \tau_\sigma^{-1}$ .)
- $S_n$  is generated by  $(1, k)$ 's (every cycle is  $\prod_{i=2}^m (1, i)$ , so cycle decomp then transposition decomp).
- $S_n = \langle (i, i+1) \text{'s} \rangle ((1n) = (n-1, n)(1, n-1)(n, n-1))$
- $S_n = \langle \{(12), (12 \dots n)\} \rangle$
- reversal:**  $a < b$  but  $\sigma(a) > \sigma(b)$ .  $R(\sigma)$  is all reversals.
- An  $m$ -cycle has exactly  $(m-1)$  reversals.

## Homomorphism

- Sign:**  $\epsilon(\sigma) = (-1)^{|R(\sigma)|}$ . even and odd permutations.
- Injective iff  $\ker = 1$ , surjective iff  $\text{im} = H$ .
- Alternating group:**  $A_n := \ker \epsilon$ .  $A_1 = S_1, |A_n| = \frac{n!}{2}$ .
- Let  $\theta: \Delta \rightarrow \Omega$  be a bijection, then  $\varphi_\theta: S_\Delta \rightarrow S_\Omega$  defined by  $\varphi_\theta(\sigma) = \theta \circ \sigma \circ \theta^{-1}$  is an isomorphism ( $\varphi_{\theta^{-1}} = \varphi_\theta^{-1}$ )  $\implies \varphi_{\theta \circ \eta} = \varphi_\theta \circ \varphi_\eta, \varphi_{\text{id}_\Delta} = \text{id}_{S_\Delta}$ .

## Subgroups

- Universal property:** if  $H \leq G$ ,  $\forall$  homomorphism  $\varphi: G \rightarrow H$ ,  $\exists!$  homomorphism  $\psi: G \rightarrow K$  where  $\text{im } \varphi \leq K \leq H$  such that  $\varphi = i_K \circ \psi$ .

- Subgroup criterion:** if  $H \subseteq G$ , then  $H \leq G$  iff  $H \neq \emptyset$  and  $xy^{-1} \in H$  for all  $x, y \in H$ .
- Every subgroup of  $\mathbb{Z}$  is  $n\mathbb{Z}$  ( $n$  is min +int in subgroup).
- Finite subgroup criterion:** if finite  $H \subseteq G$ , then  $H \leq G$  iff  $H \neq \emptyset$  and  $xy \in H$  for all  $x, y \in H$ .
- Subgroup is closed under intersection.
- Generated subgroup:**  $\langle A \rangle := \bigcap_{A \subseteq H \leq G} H$
- $\langle A \rangle$  is all products of  $A$ 's elems (use subgrp criterion).

## Cyclic groups

- Residue class:**  $\bar{a} := a + n\mathbb{Z} = \{a + kn : k \in \mathbb{Z}\}$
- Reduction-modulo- $n$ :**  $\pi(a) = \bar{a}$  uniq s.t.  $\pi(1) = \bar{1}$
- Universal property of  $\mathbb{Z}$ :**  $\forall G$  and  $\forall x \in G$ ,  $\exists!$  homomorphism  $\varphi: \mathbb{Z} \rightarrow G$  such that  $\varphi(1) = x$ . For  $\mathbb{Z}/n\mathbb{Z}$  it's analogous.
- $(x^a)^b = x^{ab}$  (let  $\psi(1) = x$  and  $\varphi_k(n) = kn$ , then  $(\psi \circ \varphi_k)(1) = x^k$ .  $(x^a)^b = (\psi \circ \varphi_a)(b) = \psi(1)^{ab}$ )
- $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \rangle$  is of order  $n$ .
- Fix  $x \in G$  and let  $\varphi: \mathbb{Z} \rightarrow G$  be the unique homomorphism with  $\varphi(1) = x$ , then  $\text{im } \varphi = \langle x \rangle \leq G$  and

$$\ker \varphi = \begin{cases} n\mathbb{Z} & \text{if } |x| = n \in \mathbb{Z}^+ \\ \{0\} & \text{otherwise} \end{cases}.$$

( $\ker \varphi \subseteq n\mathbb{Z}$  by division algo)

- $|x| = \infty \implies |\langle x \rangle| = \infty$ ;  $|x| = n \implies |\langle x \rangle| = n$ . (Let  $\varphi: \mathbb{Z} \rightarrow G$  be the homomorphism with  $\varphi(1) = x$ . Last  $x^N$  so  $x^{N+1} = x^k$  for smaller  $k$ .  $\varphi(N+1-k) = 1 \implies N+1=k$ ;  $|x| = n \implies x^{nk} = \varphi(nk) = 1 \implies x^r$ 's are pair-wise distinct)
- All infinite cyclic groups  $\cong \mathbb{Z}$
- Every subgroup of an infinite cyclic group is  $\langle x^n \rangle$  for  $n \in \mathbb{N}$  ( $\varphi$  unique isomorphism with  $\varphi(1) = x$ ,  $\psi(K) = \varphi[K]$  is bijective, every subgroup of  $\mathbb{Z}$  is  $n\mathbb{Z}$ ).
- $\forall$  group, if  $x^a = x^b = 1$ , then  $x^{\gcd(a,b)} = 1$  (by Bezout).
- $\forall$  group and  $x \in G$ ,  $\exists n \in \mathbb{Z}$  s.t.  $x^n = 1$  iff  $|x| \mid n$ .

- $\forall$  group  $G$  with  $x \in G$  s.t.  $x^n = 1$  for some  $n \in \mathbb{Z}$ . Let  $\varphi: \mathbb{Z}/n\mathbb{Z} \rightarrow G$  be the unique homomorphism s.t.  $\varphi(\bar{1}) = x$ , then  $\text{im } \varphi = \langle x \rangle \leq G$ ,  $\ker \varphi = \langle |x| \rangle \leq \mathbb{Z}/n\mathbb{Z}$
- Every finite cyclic group  $\cong \mathbb{Z}/n\mathbb{Z}$ .
- Every subgrp of an order- $n$  cyclic group is  $\langle x^a \rangle$  for  $a \mid n$
- Every subgroup of a cyclic group is cyclic.
- Let  $G := \langle x \rangle$  be a cyclic group.
  - If  $G$  is infinite, then  $G = \langle x^a \rangle$  iff  $a = \pm 1$ .
  - If  $G$  is of order  $n$ , then  $G = \langle x^a \rangle$  iff  $\gcd(a, n) = 1$ .

## Quotient groups

- $g_1 H = g_2 H \iff g_i \in g_j H \iff g_2^{-1} g_1, g_1^{-1} g_2 \in H$ .
- $|H \backslash G| = |G/H|$  ( $f: H \backslash G \rightarrow G/H$  by  $f(X) = X^{-1} := \{x^{-1} \in G : x \in X\}$ ,  $g: H/G \rightarrow H \backslash G$  by  $g(X) = X^{-1}$  is inverse)
- $H \approx gH \approx Hg$
- Lagrange's Theorem:**  $|G| = |G : H| |H|$ .
- $\langle x \rangle \leq G \implies |x| \mid |G|$  for finite  $G \implies x^{|G|} = 1$ .
- A prime-order group is cyclic.
- $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : a \leq n \text{ and } \gcd(a, n) = 1\}$  ( $\supseteq$ : Bezout  $\implies \bar{s} \bar{a} = \bar{s} a + \bar{t} n = \bar{1}$ .  $\subseteq$ :  $x = \bar{a}, y = \bar{b}$  s.t.  $xy = 1 \implies n \mid ab - 1 \implies ab - kn = 1$ )
- For any group  $G$ ,
  - $C_G(a) = N_G(\{a\})$  for all  $a \in G$ ;
  - $C_G(A) = \bigcap_{a \in A} C_G(a) \subseteq N_G(A)$ ;
  - $Z(G) = C_G(G) = \bigcap_{a \in G} C_G(a)$ ;
  - $N_G(G) = G$ ;
  - $C_G(1) = G$ .
- For every group  $G$ , the followings are equivalent:
  - $G$  is Abelian;
  - $Z(G) = G$ ;
  - $C_G(a) = G$  for all  $a \in G$ ;
  - $C_G(A) = N_G(A) = G$  for all  $A \subseteq G$ .
- $C_G(A), N_G(A) \leq G$ .
- Normal subgroup:**  $N \trianglelefteq G$  iff  $gNg^{-1} \subseteq N \forall g \in G$ .

- **Quotient group:**  $(G/N, \cdot)$  where  $N \trianglelefteq G$ . Identity is  $N$ .
- **Universal property:** Let  $N \trianglelefteq G$ . For any group  $H$  and any homomorphism  $\varphi: G \rightarrow H$  with  $N \leq \ker \varphi$ ,  $\exists! \bar{\varphi}: G/N \rightarrow H$  such that  $\varphi = \bar{\varphi} \circ \pi$  ( $\bar{\varphi}(\xi) = \varphi(g)$  iff  $\pi(g) = \xi$ ).
- **1st isomorphism thm:** For any homomorphism  $\varphi: G \rightarrow H$ ,  $\exists! \bar{\varphi}: G/\ker \varphi \rightarrow \text{im } \varphi$  s.t.  $\varphi = i_{\text{im } \varphi} \circ \bar{\varphi} \circ \pi$  is the canonical factorisation of  $\varphi$  (prove injection by  $\ker = 1$ ).
- Let  $\varphi: G \rightarrow H$  be any homomorphism, then
  - $|\text{im } \varphi| = |G: \ker \varphi|$ ;
  - if  $G$  is finite, then  $|\text{im } \varphi| \mid |G|$ ;
  - if  $G$  and  $H$  are finite, then  $|\text{im } \varphi| \mid |H|$ .
- **2nd isomorphism thm:** if  $H, K \leq G$  and  $K \trianglelefteq G$ , then  $H/(H \cap K) \cong HK/K$ . ( $K \trianglelefteq HK$ . Define  $\varphi = \pi_{HK} \circ i_H: H \rightarrow HK/K$ . Use 1st isomorphism thm to get isomorphism  $\tilde{\varphi}$ . First prove it's surjective and  $\text{im } \varphi = HK/K$ . Then  $\ker \varphi = H \cap K$  because  $hk \notin K$  for  $h \in K - H$ .)
- **3rd isomorphism thm:** if  $N, K \trianglelefteq G$  s.t.  $N \leq K$ , then  $K/N \trianglelefteq G/N$  and  $G/K \cong (G/N)/(K/N)$ . (Note:  $N \trianglelefteq K$  and  $K/N \leq G/N$  for any  $gN \in G/N$  and  $k \in K$ ,  $gkg^{-1} \in K$  and  $(gN)(kN)(g^{-1}N) = (gkg^{-1})N \in K/N$ . For each  $k$  there is  $k'$  s.t.  $k = gk'g^{-1}$  so  $kN = (gN)(k'N)(g^{-1}N) \implies K/N \trianglelefteq G/N$ . Define  $\varphi = \psi \circ \phi$  where  $\psi: G \rightarrow G/N$  and  $\phi: G/N \rightarrow (G/N)/(K/N)$  are quotient homomorphisms, then  $\varphi$  is surjective and  $\ker \varphi = K$ . Now use 1st isomorphism thm.)
- **Lattice point isomorphism thm:** Let  $N \trianglelefteq G$ ,  $\pi: G \rightarrow G/N$  be the quotient homomorphism,  $\mathcal{H}$  be the family of all subgroups of  $G$  containing  $N$  and  $\mathcal{Q}$  be the family of all subgroups of  $G/N$ , then  $\varphi: \mathcal{H} \rightarrow \mathcal{Q}$  defined by  $\varphi(H) := \pi[H]$  is a bijection such that
  1. if  $H \trianglelefteq G$ , then  $\varphi(H) \trianglelefteq G/N$ ;
  2. if  $X \trianglelefteq G/N$ , then  $\varphi^{-1}(X) \trianglelefteq G$ .

## Group actions

- $\sigma_g = a \mapsto ga$  is bijective and  $g \mapsto \sigma_g$  is a homomorphism (prove  $\sigma_{g^{-1}} = \sigma_g^{-1}$ ).

- **Action homomorphism / perm representation:**  $\varphi_\alpha(g)(a) = \alpha(g, a)$ . **Always injective**
- $\forall$  homomorphism  $\varphi: G \rightarrow S_A$ ,  $\alpha_\varphi = (g, a) \mapsto \varphi(g)(a)$  is a group action.
- $\psi: H \rightarrow G$  homomorphism,  $\varphi: G \rightarrow S_A$  perm representation,  $\alpha_\psi(h, a) = \varphi(\psi(h))(a)$  is a group action.
- **Cayley's Thm:** every group  $\cong$  some subgroup of some symmetric group.
- $\alpha \mapsto \varphi_\alpha$  is bijective (prove  $\varphi \mapsto \alpha_\varphi$  is inverse).
- if  $p$  is min prime factor of  $|G|$ ,  $|G: H| = p \implies H \trianglelefteq G$ . (consider left multiply action  $\alpha$  of  $G$  on  $G/H$ . Use 1st isomorphism thm:  $\exists K \subseteq S_{G/H}$  s.t.  $G/\ker \varphi_\alpha \cong K$ . Use Lagrange:  $|G: \ker \varphi_\alpha| \mid p!$ . Prove  $\ker \varphi_\alpha \leq H \leq G$  and  $|H: \ker \varphi_\alpha| = \frac{|G: \ker \varphi_\alpha|}{|G: H|} \mid \frac{p!}{p} = (p-1)!$ . Use Lagrange again:  $|H: \ker \varphi_\alpha| \mid |G| \implies |H: \ker \varphi_\alpha|$  has no prime factor.)
- $\forall \sigma \in S_A$ , define  $\mathcal{P}(\sigma): \mathcal{P}(A) \rightarrow \mathcal{P}(A)$  by  $\mathcal{P}(\sigma)(U) = \sigma[U]$ , then  $f(\sigma) = \mathcal{P}(\sigma)$  is a homomorphism.
- $\alpha((\sigma_A, \sigma_B), f) = \sigma_B \circ f \circ \sigma_A^{-1}$  is a group action of  $S_A \times S_B$  on  $\text{Maps}(A, B)$
- **Stable:**  $A_0 \subseteq A$  is stable under  $\alpha$  if  $\alpha[G \times A_0] \subseteq A_0$ .
- The followings are equivalent:
  1.  $A_0$  is stable under  $\alpha$ ;
  2.  $\varphi_\alpha[G] = \text{Perm}(A)_{A_0} := \{\sigma \in S_A: \sigma[A_0] = A_0\}$ ;
  3.  $\sigma_g(a) \in A_0$  for any  $g \in G$  and  $a \in A_0$ .
  4.  $\varphi_\alpha|_{G \times A_0}$  is an action group homomorphism to  $A_0$ .
- **Orbit:**  $G \cdot a$  is all  $b$  s.t.  $\exists g$  s.t.  $b = ga$ .
- **Transitive action:** only one orbit
- **Stabiliser:**  $G_a := \{g \in G: ga = a\} \subseteq G$
- $f: G/G_a \rightarrow G \cdot a$  by  $f(gG_a) = ga$  is a bijection  $\implies |G: G_a| = |G \cdot a|$
- **Fixed point:**  $ga = a \forall g \in G$ .  $A^G$  is all fixed points.
- $|A| = |A^G| + \sum |G: G_{a_i}|$  where  $G \cdot a_i$ 's non-trivial orbits

- Conjugate action: fixed points  $Z(G)$ , orbit  $C_G(g_i)$
- **$p$ -group:**  $|G| = p^\alpha$  for some  $\alpha \geq 1$ .
- **Sylow  $p$ -subgroup:** maximal  $p$ -subgroup
- Let  $|G| = p^\alpha m$ ,  $\alpha \geq 1$  and  $p \nmid m$ , then  $H \leq G$  is a Sylow  $p$ -subgroup iff  $|H| = p^\alpha$  or  $\gcd(|G: H|, p) = 1$ .
- If  $H$  is a  $p$ -group acting on  $A$ , then  $|A^H| \equiv |A| \pmod{p}$ .
- If  $H$  is a  $p$ -subgroup of a finite group  $G$ , then  $|N_G(H): H| \equiv |G: H| \pmod{p}$  (use left multiply action of  $H$  on  $G/H$  and prove  $(G/H)^H = N_G(H)/H$ ).
- Centre of any  $p$ -group is non-trivial
- **Cauchy's Thm:** For every finite group  $G$  and any prime  $p \mid |G|$ ,  $\exists g \in G$  s.t.  $|g| = p$ .
- **Sylow's Theorems:**
  1.  $\forall$  finite group  $G$  and prime factor  $p$  with multiplicity  $\alpha$  of  $|G|$ ,  $\exists$  Sylow  $p$ -subgroup of order  $p^\alpha$ .
  2.  $\forall P, Q \in \text{Syl}_p(G)$ ,  $\exists g \in G$  such that  $Q = gPg^{-1}$ .
  3.  $\forall G$  finite group with  $|G| = p^\alpha m$  with  $\alpha \geq 1$ , and  $p \nmid m \geq 1$ ,  $n_p(G) \mid m$ ;  $n_p \equiv 1 \pmod{p}$ ;  $n_p = |G: N_G(P)|$  for any Sylow  $p$ -subgroup.
- Every  $p$ -subgroup is in some Sylow  $p$ -subgroup
- **Simple group:** non-trivial and 1 and  $G$  are only normal subgroups.
- Every group of order  $< 60$  is either Abelian simple and  $\cong \mathbb{Z}/p\mathbb{Z}$  of prime order  $p$ , or not simple.
- $p$ -group with order  $> p$  is not simple (centre is normal)
- Any finite group of order  $p^\alpha q^\beta$  where  $p$  and  $q$  are primes,  $\alpha, \beta \geq 1$ , and  $p^k \not\equiv 1 \pmod{q}$  for any  $k = 1, 2, \dots, \alpha$  is not simple.